

PROVINCIAL EXAMINATION

JUNE 2024

GRADE 6

NATURAL SCIENCES AND TECHNOLOGY

TIME: 1½ hours

MARKS: 60

11 pages

DISTRICT				
NAME OF SCHOOL				
EMIS- NUMBER				
NAME AND SURNAME				
GENDER	BOY		GIRL	

SECTIONS	A					B			C			TOTAL
Questions	1	2	3	4	5	6	7	8	9	10	11	
Mark Allocation	5	5	5	5	10	5	6	10	2	3	4	60
Learner Marks												

INSTRUCTIONS

1. Answer ALL the questions in the spaces provided on this question paper.
2. Read the instructions and questions carefully before answering.
3. Write neatly and legibly.

STRANDS

NATURAL SCIENCES	MATTER AND MATERIALS
TECHNOLOGY	SYSTEMS AND CONTROL

This paper consists of **SECTION A**, **SECTION B** AND **SECTION C**.

SECTION A: LOW ORDER QUESTIONS/ COGNITIVE LEVEL 1	SECTION B: MIDDLE ORDER QUESTIONS/ COGNITIVE LEVELS 2, 3 & 4	SECTION C: HIGH ORDER QUESTIONS/COGNITIVE LEVELS 5 & 6
Q1: Water pollution, Process to purify water	Q6: Water pollution & Process to clean water	Q9: Solution as special mixtures
Q2: Soluble and insoluble substances, Rate of dissolving, Importance of wetlands	Q7: Rearranging – Scientific Investigation	Q10: Rates of dissolving
Q3: Water pollution, Clean water		
Q4: Mixtures of materials	Q8: Solids, liquids and gases, Saturated solutions, Mixtures, Solutions as special mixtures	Q11: Water pollution, Clean water
Q5: Solids, liquids and gases		
Total – 30	Total – 21	Total – 9

SECTION A

QUESTION 1: Water pollution and process to purify water

Four possible answers are given for each question below. Circle (or highlight) the letter of the correct answer.

1.1 Water pollution occurs when:

- A Harmful substances enter a water supply
 - B Industries dispose of waste safely and correctly
 - C Industries cut costs with waste disposal
 - D None of the above
- (1)

1.2 Which ONE of the following illnesses are associated with polluted water?

- A Malaria
 - B Diarrhoea
 - C Cholera
 - D COVID-19
- (1)

1.3 Water from which all germs and harmful substances have been removed, is said to be ...

- A polluted.
 - B purified.
 - C processed.
 - D pasteurised.
- (1)

1.4 Which purification method is the best to kill any germs left in the water?

- A Sieving
 - B Settlement
 - C Chlorination
 - D Filtering
- (1)

1.5 A method of cleaning water by heating it up to a very high temperature is:

- A Sieving
 - B Pasteurisation
 - C Filtering
 - D Boiling
- (1)
- [5]**

QUESTION 2

Match the statements in COLUMN B with the correct word in COLUMN A by writing the correct letter in the answer column below. **An example has already been done for you.**

COLUMN A		COLUMN B		ANSWER
E.g.	Water			e.g. F
2.1	Dissolve	A	When a dissolved substance changes into solid crystals again.	2.1
2.2	Vlei	B	An area covered by shallow water for most of the year.	2.2
2.3	Crystallisation	C	How well a substance will dissolve.	2.3
2.4	Wetland	D	An area of low flat ground that is soft and wet.	2.4
2.5	Solubility	E	When the particles of one substance spread between the particles of another substance.	2.5
		F	Chemical formulae: H ₂ O	

[5]

QUESTION 3

Use the word box below to complete the sentences below. Write your answers on the lines provided.

purification	contaminated	improve	sludge	settling	parasites
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- 3.1 When water contains germs or substances that make it unsafe to drink, we say that it is _____.
- 3.2 _____ occurs when you leave a mixture of dirty water to stand for a while, so that the heavier substances will drop to the bottom.
- 3.3 The process of cleaning dirty water is called _____.
- 3.4 The muddy substance found in water during the purification process is called _____.
- 3.5 Germs and _____ are found in dirty water.

[5]

QUESTION 4: Mixtures of materials: Labelling structures

The words below describe 5 methods of separating mixtures. Use the words in the word box below to correctly label each method of separation indicated by the picture next to each number.

Settling; Sieving; Hand sorting; Decanting; Filtering

4.1 _____



(1)

4.2 _____



(1)

4.3 _____



(1)

4.4 _____



(1)

4.5 _____

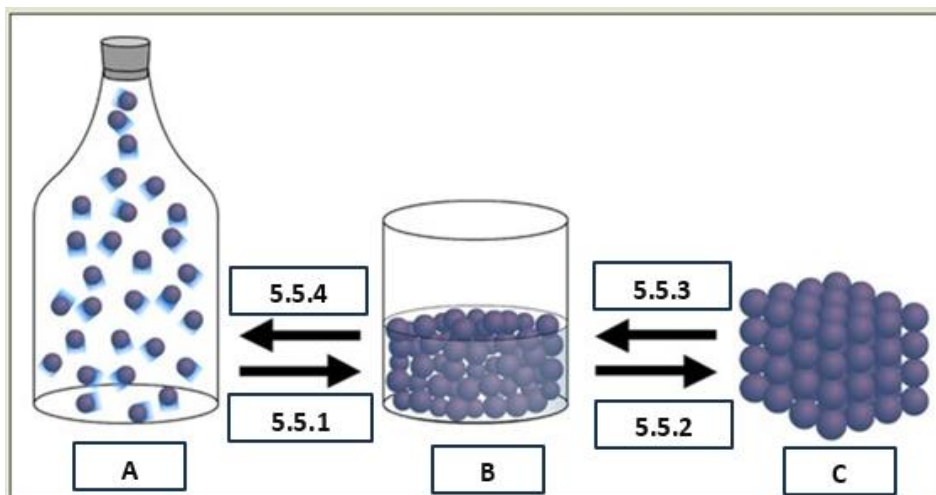


(1)

[5]

QUESTION 5

Use the illustration below on the changes of the states of matter to answer the questions that follow.



5.1 What do we call the fluid state of water? _____ (1)

5.2 What do we call the gaseous state of water? _____ (1)

5.3 What do we call the solid state of water? _____ (1)

5.4 Provide labels for the states of matter found at **A**, **B** and **C**.

A _____ B _____ C _____ (3)

5.5 Based on the diagram, what process takes place at each of the following numbers?

5.5.1 _____ (1)

5.5.2 _____ (1)

5.5.3 _____ (1)

5.5.4 _____ (1)

[10]

TOTAL SECTION A: 30

SECTION B

QUESTION 6: Water pollution and the process to clean water

Read the case study below and answer the questions that follow.

Dangerous Water

Musa Nzimakwe went to visit his family in the rural area of KwaZulu-Natal. Musa has piped water in Durban where he lives. At his rural home, people drink the water from a stream running through the village. People also used the bushes near the stream as a toilet.

After a few days, Musa became very ill with diarrhoea and had to go to the hospital. He had cholera, a disease caused by germs in water polluted with human waste. Musa was lucky, as people often die from cholera. The local municipality has now provided toilets, but they did not include an education programme to teach people how to prevent diseases such as diarrhoea and cholera. They also put in hand pumps to pump water from underground boreholes, but people still use water from the stream because they say it is too far to walk to the hand pumps.

[Adapted from an article in *EnviroTech*, May 2004, taken from *Water Wheel*, Nov/ Dec 2003, Water Research Commission].

6.1 Compare piped water to water from a stream.

6.1.1 Piped water: _____

6.1.2 Stream water: _____ (2)

6.2 Explain why the water near Musa's home is polluted.

(1)

6.3 Discuss TWO ways in which hand pumps and toilets can help prevent diarrhoea and cholera in rural communities.

6.3.1 _____

6.3.2 _____

(2)
[5]

QUESTION 7: Rearranging

Scientists always follow a very careful plan when doing investigations. There are certain steps that must be followed when doing a scientific investigation.

Rearrange the steps in the correct order by writing only the LETTER next to the correct step.

INCORRECT ORDER		CORRECT ORDER	
A	Method	Step 1:	
B	Observation and Results	Step 2:	
C	Hypothesis	Step 3:	
D	Conclusion	Step 4:	
E	Gather Materials	Step 5:	
F	Aim	Step 6:	

[6]

QUESTION 8

8.1 Explain the difference between a saturated solution and an unsaturated solution.

8.1.1 **Saturated solution** - _____

(1)

8.1.2 **Unsaturated solution** - _____

(1)

8.2 Use the table below to compare the particles in solids and gases.

Solids	Gases
(a) _____ _____ _____ _____	(a) _____ _____ _____ _____
(b) _____ _____ _____ _____	(b) _____ _____ _____ _____

(4)

8.3 8.3.1 Illustrate the arrangement of liquid particles in the frame below.



(2)

8.3.2 Explain the movement of the particles within a liquid.

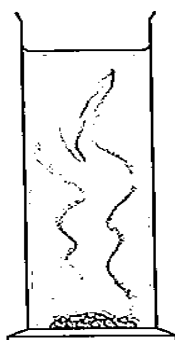
(2)

TOTAL SECTION B: 21

SECTION C

QUESTION 9: Solutions as special mixtures

Sipho added an unknown substance to water. His results are shown in the diagram below. After 10 minutes the colour was evenly spread in the water and the layer at the bottom was no longer present.

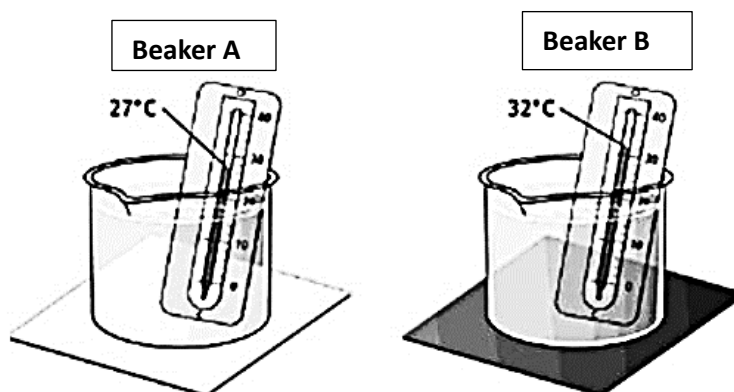


9.1 Is this an example of a solution? Give a reason for your answer.

(2)

QUESTION 10: Rates of dissolving

Hanna put the same amount of salt into these two beakers of water and stirred it well.



10.1 In which beaker will the salt dissolve quicker? Explain why the salt dissolves quicker in one beaker rather than the other beaker.

(3)

QUESTION 11

Read the article below on water pollution and answer the questions that follow.

What are the causes and effects of water pollution in Africa?

Author: Matthew McClure

Date published: 1 September 2021

What causes water pollution, and what is the effect of water pollution on at-risk communities and ecosystems in Africa?

Water pollution in Africa is being caused by humans. There's no denying it.

The very species that is so dependent on safe and clean water, is the species that is causing freshwater sources to be polluted. But, what is water pollution? Broadly defined, water pollution is the process of foreign substances entering a water body (either above or below ground) and in the process causing the water to be unusable or toxic to the ecosystem in which it is located.

The effects of water pollution are catastrophic. Plant and animal life suffers, and so do vulnerable people and communities. Hold your government to account and make sure that all Africans have access to clean, healthy water.

Make no mistake, access to clean, healthy water is a fundamental human right. Clean water is essential to life, but it is a scarce resource in Africa. The causes of water pollution and its effects on people, flora and fauna, is currently one of the biggest environmental threats to Africa. Unfortunately, water pollution in Africa is on the rise.

[Extracted from: <https://www.greenpeace.org/africa/en/blogs/49015/what-are-the-causes-and-effects-of-water-pollution-in-africa/>]

11.1 Suggest why the article states that humans are the cause of pollution.

(1)

11.2 Discuss ONE method that can be used to purify water.

(1)

11.3 Provide TWO reasons why you think it is important to keep our rivers clean.

(2)

TOTAL SECTION C: 9

TOTAL: 60

END